

Title

Stepped-Wedge Implementation of a Multicomponent Delirium Prevention Intervention in Older Surgical Patients: A Cluster Randomized Trial Protocol

Abstract

Postoperative delirium substantially increases morbidity, mortality, and healthcare costs in older populations. Effective nonpharmacologic prevention strategies exist but are underutilized in clinical practice. This stepped-wedge cluster-randomized trial will implement a structured, multidisciplinary delirium prevention program across 5 hospital systems and evaluate its effect on delirium incidence in patients aged 70 and older undergoing elective surgical procedures. The intervention combines patient education, staff training on delirium recognition, and structured daily implementation of cognitive stimulation, sensory engagement, mobility support, nutritional optimization, and sleep-promotion strategies tailored to individual risk factors. Delirium will be systematically assessed using validated protocols. Primary outcomes are postoperative delirium incidence and severity. We hypothesize the multicomponent intervention will reduce delirium occurrence by 20% or greater compared to usual care across diverse surgical settings.